AHC VISUAL INTELLIGENCE HACKATHON · 5 SEPT 2026

Real-time video anomaly detection

Can a **small** vision-language model watch live video
and report what matters — fast enough, and without crying wolf?

59.7

out of 100 · was 49.9 one hour before the end

29.8 / 35

our strongest tier — 4-minute videos localised well

2 models

a prompted VLM for appearance, a trained encoder head for duration

ARCHITECTURE

What runs, and what is new at each stage.

video ▶ 20 s windows ▶ VLM — appearance / encoder — duration ▶ merge + reshape spans ▶ alerts with timestamps

Ingest
28 videos · 5 s – 10 min

NEW
Sampled **BY TIMESTAMP**, never frame index — sources run 1.9 to 30 fps, so indices mis-time every event

WINDOWS
20 s wide, 10 s hop. Short clips get one window covering the whole clip

Appearance model
Qwen3-VL-8B · 16 frames

PROMPTING
Grouped fine-grained questions — traffic, hazard, behaviour — not one flat "what's wrong here"

OUTPUT
Constrained JSON: class, span, confidence, plain-language reason

INSTRUMENTED
Every call timed at source — p50 / p95 / max, not a config number

Duration model
SigLIP2 frozen · 8 frames

NEW
Frozen encoder → 8 vectors → **BIDIRECTIONAL GRU** reads them **IN SEQUENCE**

WHY
One frame of a person standing is normal. Eight frames of them not moving is loitering. A per-frame classifier cannot see this

Fuse & shape
CPU · no GPU

NEW · UNION
Never replace. Either model may add a class the other missed

NEW · SPAN GEOMETRY
Long videos: merge same-class events, pad ±120 s. **WORTH +6.1 MARKS ALONE**

PER-DIFFICULTY
Short clips top-2 · 4-min spans untouched · policy differs by tier, measured not assumed

Output
59.7 / 100

VALIDATED
Schema checked before every upload — ids, null timestamps at tier 1, spans inside duration

THREE STATES
no row = not run · **FAILED** = the run broke · empty = model says normal. Never blurred

■ technique we introduced ■ standard practice, applied carefully

Two models, one merge, zero retraining at the end. The last +9.8 marks came from the fuse-and-shape stage — **CPU only, no GPU, no changed class.**

WHY TWO MODELS

Over half the events came from the encoder, not the VLM.

Source of the 38 events we shipped	Events
SigLIP2 only — the VLM never saw these	**17**
Both models agreed	10
Qwen3-VL only — SigLIP never saw these	7
Derived by merging overlapping spans	4

Only **10 of 38** events had both models agreeing. The other 28 came from exactly one of them.
That is the case for running two — not a story, a count.

MODELS TESTED

Five VLMs, identical test. Two surprises.

Model	Size	Precision	Recall	Verdict
Qwen3-VL-8B-Instruct	**8B**	**0.62**	**0.40**	**Shipped**
Qwen3-VL-4B-Instruct	4B	0.77	0.50	Ties the 8B
Cosmos-Reason2-8B	8B	0.57	0.20	Half the recall of a 4B
Qwen3-VL-2B-Instruct	2B	0.30	0.15	Too small
Qwen2.5-VL-7B-Instruct	7B	0.00	0.00	Silent on 20 of 24
SigLIP2 + trained GRU head	**frozen**	**added +2.4 marks**		**Shipped**

Cosmos-Reason2-8B is purpose-built for traffic anomaly reasoning — and got half the recall of a general 4B here.
SigLIP was marked "rejected" in our own ledger, on theory. It became one of the two things that actually moved our score.

The fine-tune: we bolted on an adapter, not a new model.

LoRA on Qwen3-VL-4B

The 4-billion-parameter model stays **frozen**. We train a few small extra matrices alongside it — roughly **1% of the parameters move**. Rewriting a textbook versus adding sticky notes.

Setting	Why
rank 16	Adapter capacity. 8 is the usual default; ~2,600 examples made 16 affordable
froze the eye	Vision encoder + adapter layer frozen. Only the *reasoning* needed to change
1 epoch · 300 steps	One hour of GPU. One pass — more risks memorising 2,600 examples
lr 1e-4	10× a full fine-tune's rate, standard for LoRA — you move a small piece
batch 1 × accum 8	A 24-frame video is huge in memory. One at a time, pretending it's 8
24 frames	Default was 16 — over a 20 s window that is 0.8 fps, too coarse for a 5 s crash
save every 100	So a usable adapter existed even if the run died. It nearly did

63 minutes on an A100-80GB · loss fell to **0.124** and flattened — it converged properly

What it trained on

2,610 examples — and **71% were negatives**, windows where the right answer is *"nothing here"*.

Deliberate: sliding a 20 s window over a 4-minute video, ~90% of windows genuinely contain nothing. A model never taught to say "nothing" fires on everything.

And it lost

Replacing the baseline	**42.4**
Added to the baseline	**49.2**
Zero-shot, no fine-tune	**49.9**

Two honest reasons. It was **starved** — a 13 GB upload died unnoticed, so it got less than half its intended data. And we **over-corrected**: 71% negatives taught it to stay quiet, when our real problem was already *missing* events.

SigLIP: we never touched the model. Only what sits on top.

`8 frames → FROZEN SigLIP2 → 8 vectors → LayerNorm → bidirectional GRU (256) → Dropout → 12 classes`

A few hundred thousand parameters

Against Qwen's **4 billion**. SigLIP is a lens we never adjust — we trained a tiny brain to read what comes through it.

Setting	Why
8 frames	The GRU needs a *sequence*. Eight is enough to see "still there"
bidirectional	Reads forwards *and* backwards, so frame 3 is read knowing frame 7
20–30 epochs	Cheap — embeddings are cached, so an epoch takes seconds
dropout 0.3	Aggressive, because 660 examples overfits easily
class weights	√inverse-frequency — without it, it learns to always say "normal"
threshold 0.30	**Tuned, not default.** 0.22 cost us ~15 marks

Trained on 660 clips — of 3,173 available

Not a design choice. The full dataset never uploaded, so we built a class-balanced subset that fit the time we had. **55 per class.**

~68% validation accuracy — a real number, best epoch kept, 10% held out.

> The **first** SigLIP run reported **"100% validation accuracy"**. Meaningless — 77 embeddings, validated on **7 samples**, because an upload had died unnoticed. One of our ten silent failures.

It earned its place

17 of our 38 shipped events — more than the VLM contributed. And it found **loitering**, a class we score 0/6 on that no VLM proposed all day.

> It saw a fifth of the available data and still contributed **more events than the 8-billion-parameter VLM.** The cheap thing outperformed the expensive thing.

THE BREAKTHROUGH · FINAL HOUR

+9.8 marks. We changed no model.

Eight prediction sets — three model sizes, five window configs, a fine-tune, targeted prompts — all scored **exactly the same** on the long videos.

That invariance was the clue. If changing the model changes nothing, the failure isn't in the model.

Our predicted spans were **1–8 seconds** inside videos of **327–602 seconds**. A 5-second span can never overlap a 100-second event by half — **geometrically impossible, however right the class is.**

49.9 → 56.0

widening spans to 112–138 s.
No new inference. No changed class.

Our classes had been right all along.

→ 59.7

adding SigLIP2, which predicts

loitering

— a class we score 0/6 on and

no VLM proposed all day.

> The evidence was in our own notes from that morning: the sample data held a **125-second** event. We wrote down that these events are long — then spent hours swapping models instead of spans.

HOW WE WORKED

Three AI agents, one repository.

Agent	Role	Owned
Coordinator	Research, strategy, every GPU run, every commit	inference lane, Modal driver
Builder	Data pipeline, synthetic video generation, training, scoring	data lane, tests
Comms	Live demo, slides, process record	demo, docs

- **Ownership by directory.** No agent edits another's files. We learned it the hard way — two agents built the same four modules before the rule existed.
- **A written ledger, not chat.** Every technique marked done / running / rejected — **with the reason**. An agent could crash, restart and resume from files. It happened twice.
- **They corrected each other.** One caught a scorer using the wrong matching rule. One caught a run that exited "successfully" having done 6 of 34 videos.